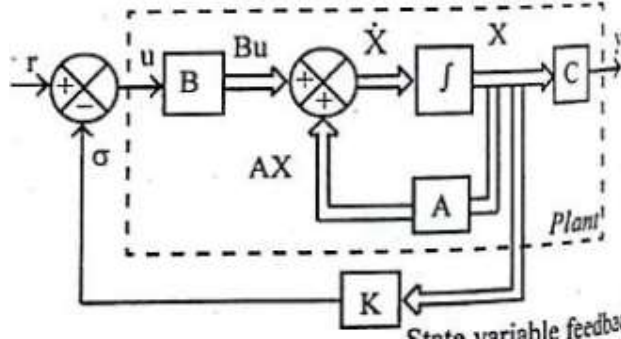


DRAFT Scheme of Valuation/Answer Key			
(Scheme of evaluation (marks in brackets) and answers of problems/key)			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
SEVENTH SEMESTER B. TECH(S,FE) DEGREE EXAMINATION, MAY 2024 (2019 Scheme)			
Course Code: EET401			
Course Name: ADVANCED CONTROL SYSTEMS			
Max. Marks: 100			Duration: 3 Hours
PART A			
		Answer all questions, each carries 3 marks.	Marks
1		$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -3 & -5 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \quad [A] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -3 & -5 & -4 \end{bmatrix}$	(3)
2		$\dot{x} = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u$ $y = [0 \quad 0.5]x$	(3)
3		Theorem-1.5, Example-1.5	(3)
4		$TF = C[SI - A]^{-1} B + D$	(3)
5		Canonical model-1 Conditions-2	(3)
6			(3)
7		Output contains only the fundamental component, non-linearity have symmetry, and the linear part of the systems have low pass filter characteristics(1+1+1)	(3)
8		Derivation- 2 Waveforms- 1	(3)
9		Difficult to come up with suitable Lyapunov function , conditional stability (1.5+1.5)	(3)
10		Sylvester's criteria for positive and negative definiteness- 1.5 marks each	(3)
PART B			

Answer one full question from each module, each carries 14 marks.			
		Module I	
11	a)	$\dot{x} = \begin{bmatrix} \frac{-12}{5} & \frac{6}{5} \\ -3 & \frac{-6}{5} \end{bmatrix} x + \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} u$	(9)
	b)	$\dot{x} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u$, $y = \begin{bmatrix} 40 \\ 3 & -15 & 5 \\ 3 \end{bmatrix} x$	(5)
		OR	
12	a)	$\dot{x} = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} u$, $y = [1 \ 0 \ 0]x$	(7)
	b)	diagonalised matrix $A = \begin{bmatrix} -1 & 0 \\ 0 & 5 \end{bmatrix}$	(7)
		Module II	
13	a)	STM = $\begin{bmatrix} e^{-t} & e^{-t}-e^{-2t} \\ 0 & e^{-2t} \end{bmatrix}$ (3marks) $y(t) = -\frac{1}{2} + \frac{1}{2}e^{-2t} - 2e^{-2t}$ (4marks)	(7)
	b)	STM = $\begin{bmatrix} -3^k & 0 \\ 0 & -2^k \end{bmatrix}$ (3marks) $y(k) = \frac{1}{4}(-3)^k - \frac{1}{3}(-2)^k + \frac{1}{12}u(k)$ (4marks)	(7)
		OR	
14	a)	$TF = \frac{2s+1}{s^3+3s^2+2s}$	(8)
	b)	Derivation of the solution of homogeneous state equation-4 marks What is state transition matrix?- 2 marks	(6)
		Module III	
15	a)	$Q_c = \begin{bmatrix} 0 & 0 & a_1 a_2 \\ 0 & a_2 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ $ Q_c = -a_1 a_2^2$ For Q_c to be nonsingular $ Q_c \neq 0$ $\therefore a_1 \neq 0$ and $a_2 \neq 0$	(7)
	b)	State model-2 Controllability check-1, $u = -[1 \ \sqrt{2}]x - 4$ marks	(7)
		OR	
16	a)	$\dot{x} = \begin{bmatrix} -3 & -1 & 0 \\ 2 & 0 & 0 \\ 0 & -1 & -1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} u$, $y = [1 \ 0 \ 1]x + u$	(8)

		Controllable and not observable State model (2 marks), controllability(3marks), Observability(3marks)	
	b)	$K_e = \begin{bmatrix} 16 \\ 84.6 \end{bmatrix}$	(6)
		Module IV	
17	a)	Polar plot- 4 marks $-\frac{1}{K_N}$ plot- 4 Marks System is stable – 2 marks	(10)
	b)	2 types- 2, examples- 2	(4)
		OR	
18	a)	Schematics- 2, derivation steps -5 , final equation- 2 marks	(9)
	b)	Stability conditions-3 marks , stable and unstable limit cycle- 2 marks	(5)
		Module V	
19	a)	Stability and instability definitions- 2marks each	(4)
	b)	Singular points-2 marks, plot- 8 marks (Minimum 4 points)	(10)
		OR	
20	a)	Select lypanuv function – 2 marks state model- 2 marks stability- 4 marks $V(x) = \frac{1}{2}x_2^2 + \frac{1}{2}Kx_1^2$ and $\dot{V}(x) = -Bx_2^2$ Stable for $B > 0$	(8)
	b)	Steps-2 Stability – 4 marsk Stable for $(1 - 2x_1x_2) > 0$	(6)
